VRYFD

Verified

High-Level Documentation

Decentralized Event Attendance & Certification Platform

Network: Ethereum (Sepolia Testnet)

Stack: Solidity | Next.js | Node.js | IPFS

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Table of Contents

1. Project Overview
2. Problem Statement
3. Solution
4. System Architecture
5. Core Features
6. User Workflow
7. Technology Stack
8. Smart Contract Architecture
9. Backend Layer
10. Frontend Layer
11. Token Economy
12. Data Architecture
13. Key Design Patterns
14. Security Model
15. Benefits & Use Cases
16. Future Scope

1. Project Overview

VRYFD (pronounced *Verified*) is a decentralized event attendance and certification platform built on the Ethereum blockchain. It replaces traditional paper certificates and easily editable PDFs with tamper-proof, blockchain-anchored digital credentials that anyone can verify in seconds.

The platform runs on the **Sepolia testnet**, meaning all interactions use free test ETH with zero monetary risk. Every check-in, badge, certificate, and token reward is recorded on-chain, creating a permanent and publicly auditable record of participation.

Core Value Proposition

VRYFD transforms event attendance from an easily forged claim into a cryptographically verifiable fact. Attendees build a portable, trustless credential portfolio. Organizers eliminate fake attendance. Employers and institutions can verify claims instantly via QR code.

2. Problem Statement

Current event participation systems suffer from fundamental trust and efficiency problems:

- **Certificate Fraud** - PDF and image-based certificates can be trivially forged or edited, undermining the credibility of legitimate achievements.
- **Verification Bottleneck** - Employers and institutions have no reliable, automated way to confirm whether a candidate actually attended an event or earned a credential.
- **Manual Inefficiency** - Traditional attendance systems rely on sign-in sheets, manual headcounts, or honor-based reporting, all of which are error-prone and exploitable.
- **No Portability** - Credentials are siloed within individual platforms and cannot be independently verified outside the issuing organization's ecosystem.

3. Solution

VRYFD addresses these challenges through a blockchain-native approach:

- **Soulbound digital badges** (non-transferable NFTs) that serve as tamper-proof proof of attendance
- **On-chain attendance records** creating permanent, immutable participation history
- **Public verification via QR codes** enabling anyone to confirm credential authenticity in seconds
- **Gamified engagement** through milestone NFT rewards and a platform token ($VRYF)
- **ENS identity integration** replacing cryptic wallet addresses with human-readable names
- **Gasless check-in** via EIP-712 signatures so attendees never pay transaction fees

4. System Architecture

VRYFD follows a three-layer architecture with the blockchain as the ultimate source of truth:

Layer	Technology	Responsibility
Smart Contracts (On-Chain)	Solidity ERC-1155 + ERC-20	Attendance records, soulbound badges, milestone NFTs, certificates, VRYF tokens
Backend Relayer	Node.js / Express	Gas-paying relayer, join code validation, EIP-712 signature verification, IPFS uploads
Frontend	Next.js (TypeScript) wagmi + RainbowKit	Wallet connection, event browsing, check-in UI, badge/cert management
Storage	IPFS via Pinata JSON file cache	Badge/certificate metadata & images, server-side state persistence

The backend acts as a **trusted relayer** holding ORGANIZER_ROLE and RELAYER_ROLE on the smart contract. It pays gas on behalf of users for check-ins and confirmations, ensuring a frictionless attendee experience. All critical state ultimately lives on-chain.

5. Core Features

5.1 ENS-Verified Identity

VRYFD integrates with the Ethereum Name Service to replace wallet addresses with human-readable names (e.g., `alice.eth`). ENS names are resolved against Ethereum mainnet and displayed on leaderboards, certificates, and attendance records. ENS is optional; all features work with a plain wallet address.

5.2 Soulbound POAP Badges

Each confirmed attendance earns the attendee a Proof of Attendance Protocol (POAP) badge implemented as a soulbound ERC-1155 token. These badges cannot be sold, transferred, or given away, ensuring they represent genuine personal participation. Badge metadata (event name, date, image) is stored on IPFS for permanence even if the platform goes offline.

5.3 Milestone NFT Rewards

A three-tier gamification system rewards consistent engagement with collectible milestone NFTs:

Tier	Threshold	Reward
Bronze Explorer	5 events	Bronze NFT + 100 VRYF
Silver Pioneer	10 events	Silver NFT + 250 VRYF
Gold Legend	20 events	Gold NFT + 500 VRYF

Milestone NFTs are also soulbound and claimed manually by the attendee when their event count crosses each threshold.

5.4 Blockchain Certificates

After an event ends, organizers issue time-stamped certificates anchored to the blockchain. Each certificate contains a unique ID, the attendee's wallet/ENS, an issue timestamp, and a QR code linking to a public verification page. Certificates can be shared on LinkedIn, downloaded as PNGs, or verified by scanning the QR code. Organizers can revoke certificates if issued in error, and revocation status is publicly visible.

6. User Workflow

6.1 End-to-End Attendee Flow

Step	Action	Details
1	Connect Wallet	MetaMask or WalletConnect on Sepolia testnet
2	Enter Join Code	6-character code displayed by organizer at the venue
3	Organizer Confirms	Organizer verifies physical presence and confirms on-chain
4	Claim POAP Badge	Soulbound ERC-1155 minted to wallet (+25 VRYF)
5	Earn Milestones	Bronze/Silver/Gold NFTs unlocked at 5/10/20 events
6	Receive Certificate	Organizer issues time-stamped, QR-verifiable certificate
7	Share & Verify	Download, post on LinkedIn, or scan QR for public verification

6.2 Organizer Flow

Step	Action	Details
1	Create Event	Name, emoji, color, start/end dates via organizer dashboard
2	Deploy On-Chain	Smart contract records event; join code generated automatically
3	Share Join Code	Display at venue via slides, whiteboard, or printed cards
4	Confirm Attendees	Review pending list; confirm individually or batch-confirm all
5	Issue Certificates	Bulk-issue to all confirmed attendees after event ends
6	Manage & Revoke	Edit events mid-flight; revoke certificates if needed

6.3 Verification Flow

Any third party (employer, school, institution) can verify a VRYFD certificate without an account or wallet. They scan the certificate's QR code, which opens a public verification page reading directly from the smart contract. The page displays event name, attendee identity, timestamp, and valid/revoked status, with a link to the raw Etherscan transaction.

7. Technology Stack

Category	Technology	Purpose
Blockchain	Ethereum (Sepolia Testnet)	On-chain state and transaction execution
Smart Contracts	Solidity (ERC-1155 + ERC-20)	Attendance, badges, milestones, certs, tokens
Development Framework	Hardhat	Contract compilation, testing, and deployment
Backend	Node.js / Express	Relayer service, API routes, IPFS uploads
Signatures	EIP-712 Typed Data	Gasless, replay-proof check-in signatures
Frontend	Next.js 13+ (TypeScript)	Server-side rendering, app router
Wallet Integration	wagmi + viem + RainbowKit	Wallet connection, on-chain reads/writes
Decentralized Storage	IPFS via Pinata	Badge/certificate metadata and images
Blockchain Interaction	ethers.js	Contract calls from backend relayer
Identity	ENS (Ethereum Name Service)	Human-readable wallet identities
Block Explorer	Etherscan (Sepolia)	Transaction verification and auditing

8. Smart Contract Architecture

VRYFD uses two primary smart contracts deployed on Sepolia:

8.1 VryfdAttendance.sol

The core contract handling events, attendance records, soulbound badges, milestone NFTs, and certificate issuance. It implements ERC-1155 for multi-token support with a custom `_update()` override that blocks all transfers except minting and burning, enforcing the soulbound property.

Role-Based Access Control

Role	Capabilities	Held By
ORGANIZER_ROLE	Create events, confirm attendance, issue/revoke certificates, set milestone URIs	Relayer wallet
RELAYER_ROLE	Submit gasless check-ins, batch confirmations	Relayer wallet

Key Contract Functions

Category	Function	Description
Events	createEvent()	Creates event + soulbound badge token ID
Attendance	checkInFor()	Organizer confirms physical presence on-chain
Attendance	confirmAttendanceGasless()	Relayer submits EIP-712 signed check-in
Attendance	batchConfirmAttendance()	Batch version for multiple attendees
Badges	claimBadge()	Mints 1x soulbound ERC-1155 POAP (+25 VRYF)
Milestones	milestoneReward()	Mints soulbound NFT per unlocked tier
Certificates	issueCertificate()	Stores Certificate struct on-chain
Certificates	revokeCertificate()	Sets isValid = false
Certificates	verifyCertificate()	Public view: event, attendee, timestamp, validity

8.2 VRYFToken (ERC-20)

A standard ERC-20 token with a hard cap of 1,000,000 VRYF. Only the VryfdAttendance contract can mint new tokens, ensuring supply is exclusively driven by genuine platform engagement. On Sepolia, VRYF has no monetary value and serves as a participation score.

9. Backend Layer

The backend is an Express.js server functioning as a **trusted relayer**. It holds on-chain roles (ORGANIZER_ROLE + RELAYER_ROLE) and pays gas on behalf of users, enabling a gasless experience for attendees. There is no traditional database; server-side state is persisted in JSON files.

Key Services

- **relayerService.js** - Wallet initialization, RPC setup with fallback, join code validation, EIP-712 signature verification, attendee confirmation, and bulk certificate issuance.
- **certificateService.js** - Generates 1200x800 PNG certificates with QR codes, uploads images and ERC-1155 metadata JSON to IPFS via Pinata.
- **ipfsService.js** - Pinata API wrappers for pinning JSON and files; builds badge, milestone, and certificate metadata.

API Routes

Method	Route	Purpose
POST	/api/relay/createEvent	Create event on-chain (relayer pays gas)
POST	/api/relay/joinCheckin	Validate join code + signature; add to pending
POST	/api/relay/confirmAttendee	Confirm single attendee on-chain
POST	/api/relay/confirmAllAttendees	Batch confirm all pending attendees
POST	/api/relay/issueCertificatesBulk	Issue certs to all confirmed attendees
GET	/api/certificates/mine/:address	Fetch on-chain certificates for a wallet
POST	/api/certificates/generate	Generate PNG + QR + upload to IPFS
GET	/api/verify/:certId	Public certificate verification endpoint

10. Frontend Layer

The frontend is a Next.js 13+ application using TypeScript with the App Router. Wallet connectivity is handled by wagmi and RainbowKit, providing a polished connect modal supporting MetaMask, WalletConnect, and other popular wallets.

Pages & Components

Page	Description
Dashboard	Hero section + 5 stat cards (Active Events, Check-ins, Badges, Tier, VRYF Balance) + events grid
EventsPage	Main check-in interface: event list (left) + 3-part check-in flow (right) with join code, confirmation status, a
BadgesPage	POAP badge grid (soulbound ERC-1155) + milestone section with 3-tier progress bars and claim button
CertificatePage	On-chain certificate list with per-cert actions: verify, download QR, share on LinkedIn
OrganizerPage	5-tab organizer dashboard: Create, Deploy, Manage, Attendance, Certificates
LeaderboardPage	Ranked by attendance count; shows ENS, event count, tier, and VRYF balance

11. Token Economy - $VRYF

VRYFD includes a utility token ($VRYF) that serves as a participation score and engagement metric. It is earned automatically through platform activity with no purchases or staking.

Action	VRYF Reward	NFT Minted
Event check-in confirmed	+10 VRYF	-
POAP badge claimed	+25 VRYF	Soulbound ERC-1155
Bronze milestone (5 events)	+100 VRYF	Soulbound NFT
Silver milestone (10 events)	+250 VRYF	Soulbound NFT
Gold milestone (20 events)	+500 VRYF	Soulbound NFT

Token Properties

Standard: ERC-20 | Max Supply: 1,000,000 VRYF | Network: Sepolia Testnet | Monetary Value: None (testnet) | Minting: Only by VryfdAttendance contract | Visibility: MetaMask (import token address), Dashboard, Leaderboard

12. Data Architecture

VRYFD uses a three-tier data model where on-chain state is the source of truth, the backend maintains operational caches, and the frontend provides per-wallet local caching:

Layer	Storage	Data
On-Chain (Source of Truth)	Sepolia blockchain	AttendanceRecord per (eventId, attendee), Certificate structs, ERC-1155 balances, VRYF ERC-20 balances
Backend Cache	JSON files (backend/data/)	deployedEvents.json (tx hashes, join codes, confirmed counts), events.json, pending.json
Frontend Cache	localStorage	Per-wallet: check-ins, badges, VRYF balance, claimed milestones, badge mint times

13. Key Design Patterns

Gas Efficiency

- **Relayer pattern** - Backend wallet pays all infrastructure gas costs, removing cost barriers for attendees
- **Gasless check-in** - Users sign EIP-712 typed data (free); the relayer submits the on-chain transaction
- **Batch operations** - confirmAllAttendees() and issueCertificatesBulk() reduce transaction count

Scalability

- **Paginated log queries** - Breaks blockchain event scans into 500-block chunks for RPC compatibility
- **In-memory caching** - Confirmed attendees cached server-side to minimize repeated chain reads
- **RPC failover** - Automatic switching between Alchemy and public endpoints on 429 rate limits

Data Integrity

- **Soulbound enforcement** - _update() override blocks all token transfers (mint/burn only)
- **Replay-proof signatures** - EIP-712 with per-wallet nonces stored on-chain
- **Duplicate prevention** - issueCertificatesBulk() reads existing certs before issuing new ones
- **Public verification** - /api/verify/:certId is read-only with no authentication required

14. Security Model

VRYFD employs multiple security layers to prevent attendance fraud and credential forgery:

- **Two-party check-in** - Attendees submit a venue-specific join code, then the organizer confirms physical presence. Both steps are required, making remote fraud nearly impossible.
- **Soulbound tokens** - POAP badges and milestone NFTs are non-transferable at the smart contract level, preventing credential trading or delegation.
- **EIP-712 typed signatures** - Check-in messages are signed with domain-specific typing and per-wallet nonces, preventing replay attacks and cross-contract exploitation.
- **Role-based access control** - Only wallets with ORGANIZER_ROLE can create events, confirm attendance, and issue or revoke certificates.
- **On-chain immutability** - All records are permanently stored on the blockchain. Certificates can be revoked (marked invalid) but never deleted from history.
- **IPFS-pinned metadata** - Badge and certificate metadata is stored on IPFS, ensuring it survives even if the VRYFD frontend or backend goes offline.
- **Public verification** - Anyone can verify credentials without an account by scanning a QR code or querying the smart contract directly via Etherscan.

15. Benefits & Use Cases

For Attendees & Students

- Build a verifiable, portable credential portfolio linked to your blockchain identity
- Showcase genuine achievements on LinkedIn with QR-verifiable certificates
- Earn progression rewards (milestone NFTs and VRYF tokens) for consistent engagement

For Event Organizers

- Eliminate fake attendance and automate the certification process
- Batch-confirm attendees and bulk-issue certificates with minimal effort
- Track real-time attendance with on-chain data integrity

For Employers & Institutions

- Instantly verify candidate claims by scanning a QR code or querying Etherscan
- Eliminate manual verification workflows with cryptographically anchored proof
- Cross-reference attendance history across multiple events and organizers

16. Future Scope

VRYFD's roadmap includes several planned enhancements to expand platform capabilities:

- **Job platform integration** - Direct credential sharing with hiring platforms for streamlined background verification
- **Advanced analytics dashboard** - Detailed engagement metrics, attendance trends, and organizer insights
- **DAO-based event governance** - Community-driven event curation and quality control via decentralized governance
- **Cross-chain compatibility** - Expand beyond Ethereum to support Polygon, Arbitrum, Base, and other L2 networks
- **Mainnet deployment** - Transition from Sepolia testnet to Ethereum mainnet for production use
- **Mobile native app** - Dedicated iOS and Android applications with push notifications and offline capabilities
- **CSV/API export** - Built-in attendee list export for organizers without requiring direct contract queries

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VRYFD transforms event attendance verification from an easily forged claim into a cryptographically anchored, publicly verifiable fact. By combining soulbound NFTs, blockchain certificates, ENS identity, and gamified rewards on a gasless infrastructure, it bridges the gap between education, events, and professional credibility.

VRYFD - Verifiable Real-World Engagement via Decentralized Proof